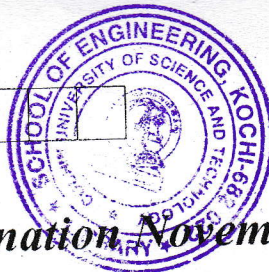


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# **B.Tech. Degree III Semester Examination November 2018**

## **CS 15-1302 LOGIC DESIGN** (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

### **PART A** (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Perform  $(845)_8 - (375)_8$ .  
 (b) Convert  $(556)_{10}$  to binary and to octal.  
 (c) Design half adder and full adder using NOR gates.  
 (d) Find the complement of  $A^1B^1 + CD$  and reduce it to a minimum number of literals.  
 (e) What is Johnson counter? Explain with a diagram.  
 (f) Design a  $4 \times 16$  decoder using two  $3 \times 8$  decoders.  
 (g) Differentiate between PLA and PAL.  
 (h) Differentiate between combinational and sequential circuits.  
 (i) Explain how TTL to CMOS interface can be done.  
 (j) What is RTL and DTL?

### **PART B**

(4 × 10 = 40)

- II. Simplify the following expression using K map method.  
 (a)  $F(A, B, C, D, E) = (0, 2, 4, 6, 9, 13, 21, 23, 24, 25, 26, 29, 31)$  (5)  
 $d(A, B, C, D, E) = (8, 10, 16, 18)$ .  
 (b)  $F = ACE + A^1CD^1E + A^1C^1DE$   $d = DE^1 + A^1D^1E + AD^1E^1$ . (5)
- III. (a) What are universal gates? With necessary expressions implement NOT, AND, OR and XOR using NAND gates only. (5)  
 (b) Using Boolean laws, show that  $AB + A^1C + BC = AB + A^1C$ . (5)
- IV. Implement the function using a multiplexer  $F(A, B, C, D) = (1, 3, 4, 11, 12, 13, 14, 15)$ . (10)
- V. Design a 4-bit carry look ahead adder and explain the operation. What is the number of gate delays involved in the circuit? (10)
- VI. Design a mod-9 synchronous down counter using JK flip-flop. (10)
- VII. Explain with diagram the working of (i) ripple counter (ii) ring counter (iii) Synchronous counter. (10)
- VIII. (a) Explain with the circuit diagram TTL NAND gate. (6)  
 (b) Define the terms: (i) Fan out (ii) Noise Margin (iii) Propagation Delay (4)  
 (iv) Power dissipation.
- IX. Describe in detail, how can we interface the CMOS and TTL logic family devices with the help of figures. (10)